

Algorithmic Decision-Making: When Machines Reason for Organizations

Executive Summary

Algorithmic decision-making transfers cognitive tasks from human minds to computational systems. Under Active Inference, algorithms are inference engines with generative models — but unlike human models, algorithmic models are explicit, auditable, and often opaque in different ways. This module examines algorithmic decision support, AI governance, explainability, the ethics of automated inference, and how organizations maintain accountability when machines make decisions.

Learning Objectives

1. Classify **algorithmic decision-making** by autonomy level and decision impact
 2. Design **AI governance frameworks** that maintain accountability for automated decisions
 3. Understand **explainability and interpretability** challenges in ML-based inference
 4. Apply **ethical reasoning** to automated decision systems
 5. Navigate the **organizational politics** of algorithmic authority
-

Key Concepts

1. Levels of Algorithmic Authority

Level	Human Role	Algorithm Role	Example
Decision support	Human decides, algorithm informs	Provides analysis and recommendations	Business intelligence dashboards
Decision augmentation	Human decides with algorithmic input	Co-equal partner in inference	Diagnostic decision support in medicine
Decision automation	Human monitors, algorithm decides	Full authority within bounds	Algorithmic trading, automated hiring screening
Autonomous systems	Human designs only	Self-directed inference and action	Autonomous vehicles, real-time ad bidding

2. AI Governance

Governance Element	Purpose	Implementation
Accountability	Who is responsible when the algorithm errs?	Clear ownership of each automated decision domain
Transparency	Can we explain why a decision was made?	Logging, audit trails, explainability methods
Fairness	Does the algorithm treat people equitably?	Bias testing, fairness metrics, impact assessment
Oversight	Is there human review of algorithmic decisions?	Escalation procedures, periodic audits, kill switches

3. Explainability Challenge

Case Study — Criminal Justice Risk Scores: COMPAS (Correctional Offender Management Profiling) uses algorithmic risk scoring to inform sentencing and parole decisions. ProPublica's investigation revealed racial disparities in the algorithm's predictions. The case illustrates the governance challenge: when algorithms make consequential decisions about people's lives, explainability, fairness, and accountability are not optional — they are fundamental requirements. The algorithm's generative model must be interrogable.

4. Organizational Politics of Algorithms

Introducing algorithmic decision-making redistributes power within organizations. When algorithms replace human judgment, those whose expertise is displaced may resist. Successful implementation requires:

- Involving affected stakeholders in the design process
- Clearly communicating what the algorithm does and doesn't decide
- Preserving human override capability for consequential decisions
- Building organizational capability to evaluate algorithmic performance

Cross-References

- For organizational cognition, see [Organizational Systems: Cognition](#)
 - For team sensemaking, see [Collective Intelligence: Cognition](#)
 - For strategic reasoning, see [Strategic Modeling: Cognition](#)
-

Summary

Concept	Digital Transformation Meaning
Algorithmic decision-making	Transferring inference tasks from humans to computational systems
AI governance	Framework ensuring accountability, transparency, fairness, and oversight
Explainability	The ability to articulate why an algorithm made a specific decision
Fairness	Ensuring algorithmic decisions don't systematically disadvantage groups
Organizational politics	Power redistribution when algorithm authority replaces human authority

References

- O'Neil, C. (2016). *Weapons of Math Destruction*. Crown.
- Mittelstadt, B. D. et al. (2016). The ethics of algorithms. *Big Data & Society*, 3(2).